

A Table-driven Li-Ion Battery Model for a BMS Development Platform – Modeling, Measurements, Implementation and Validation

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Abstract

In this paper a simulation environment for the development of battery management systems (BMS) with respect to Lithium-ion cells is presented. This platform should allow for the verification of the suitability and of the interaction between selected hardware components (Li-Ion batteries and semiconductor circuits for battery monitoring), and of the software of a battery management system (BMS) by simulation. The necessary mathematical models for an equivalent circuit model (ECM) based on the theory of passive electric elements are developed. It contains a general system of nonlinear ordinary differential equations to describe the dynamical behavior of Lithium-ion with respect to internal voltages U_i , $i = 1, \dots, n$, temperature and the state of charge SOC and the cell temperature T . The parametrization of the cells was carried out on the base of CIT (Current Interruption Technique) measurements in an Accelerating Rate Calorimeter (ARC) at different temperatures T and SOC in the time domain with a Least-Squares approach and subsequent optimization. From the parametrization look-up tables are created which are needed for the model implementation in the software. The models have been implemented in the modeling language SystemC AMS using the lookup-tables. The validation was performed with a current profile for an electric vehicle based on a NEDC (New European Driving Cycle) velocity profile.

Keywords: Battery Management System (BMS), Equivalent circuit model (ECM), Current interruption technique (CIT), Look-up table (LUT), Accelerating Rate Calorimeter (ARC)

1 Introduction

Li-ion batteries are widely used in today's electric vehicles (EV), hybrid electric vehicles (HEV), cell phones and many other industrial applications. Very often it is necessary to control the battery with a Battery Management System

(BMS) to provide useful and important information about the battery system like the State of charge (SOC) to predict the range of an EV/HEV or the temperature to avoid thermal runaway as the worst case scenario in battery applications. Therefore accurate models and simulations are of high importance. Various models with different complexities are possible. The range of such

models varies from the fully coupled electrochemical thermal model describing the dynamics of the battery in terms of partial differential equations (PDE) over Reduced Order Models (ROM) as simplification of the PDE model up to Equivalent Circuit Models (ECM) which use a phenomenological description of the electrical behavior in terms of ordinary differential equations [1]. Starting with the electrochemical-thermal model, see [2], [3] and the references therein, a spatio-temporal description of the concentration of the Li-ions and potentials in the electrodes as well as the temperature distribution of the battery is given in terms of a multi-scale, multiphysics system of elliptic and parabolic differential equations using homogenization methods. This model is able to capture almost all physical relevant phenomena at the expense of corresponding complexity with respect to spatial and temporal resolution. As a consequence it cannot be applied in BMS today due to its high complexity. A possible ansatz to reduce this complexity is the use of Reduced Order Models (ROM) which can be derived from the previous model using some simplification assumptions like the eigenfunction expansion method [4]. One possible candidate is the Single Particle Model (SPM) which is able to reproduce the dynamics of the original model in some ranges with good agreement and with less complexity. The corresponding partial differential equations can be converted into ordinary differential equations by the methods of lines (MOL) to reach less complexity. But these equations are also complex in comparison to the Equivalent Circuit Models (ECM).

An ECM model describes the pure electrical behavior of the battery in terms of voltages, currents, resistances and capacitances. Using the theory of passive electrical networks the resulting equations are a low dimensional system of ordinary differential equations which is in general nonlinear, since the system parameter depends on temperature T and state of charge SOC . Due to these dependencies an extended ECM model is used together with an additional differential equation for the SOC and a simplified temperature model in form of a linear inhomogeneous differential equation. These models are able to capture the essential electrical phenomena with good accuracy and less complexity. An overview of ECMs can be found in [5, 6]. They are easy to embed in BMS for real time application to ensure safe and

reliable operation conditions. Therefore in the project IKEBA "Integrated components and integrated design of energy efficient battery systems" a simulation environment has been introduced for the development of battery management systems using ECM for battery modelling [7]. This virtual environment shown in Fig.1 enables the analysis of interactions of battery cells as well as the interaction of different integrated circuits for monitoring cell voltages, cell temperatures and charge balancing. Furthermore the testing of the battery management system software for different load profiles is also possible. The system is able to investigate the nominal case and furthermore the faulty behavior of individual components can also be taken into account. As modeling languages SystemC [8] and SystemC AMS [9, 10] are used. The main purpose is, due to the high speed of the simulations to examine the largest possible number of test cases. This plays a central role in the context to provide accurate and fast to be evaluated models for the battery cells.

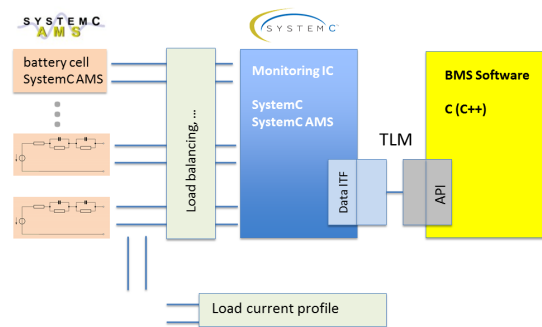


Figure1: Principle of the virtual design platform for BMS development.

This paper is organized as follows: In section 2 the parametrization of the battery cell model will be described based on CIT (Current Interruption Technique) measurements in an Accelerating Rate Calorimeter (ARC) at different temperatures T and SOC . Section 3 demonstrates the validation of the virtual prototype of the simulation environment using the NEDC (New European Driving Cycle) velocity profile.

2 Parametrization of the cell model

The first basic step is the determination of the equivalent circuit model (ECM) for the battery cell. Based on experimental measurements in an ARC, a general model has been used including a voltage source for the open circuit voltage, an

Ohmic resistor for the inner resistance of the cell and two parallel RC -branches describing various electro-chemical mechanisms (s. Fig.2). A similar second order model (dual polarization model) has been used e.g. by Einhorn et al. [11] or Daowd et al [12]. In the latter work the parameters were estimated for different SoC, temperatures and C -rates using a built in tool in Simulink®.

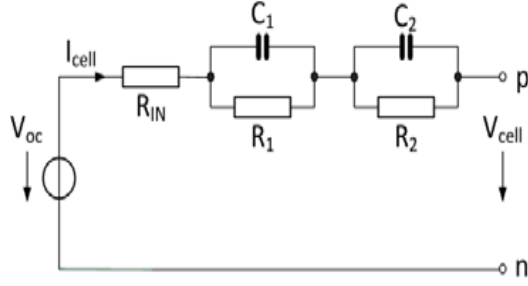


Figure2: General scheme of 2nd order ECM model.

In general it is assumed that the parameters of the second order model depend on the state of charge SOC and the cell temperature T .

$$\frac{dU_i}{dt} = \left(-\frac{1}{R_i C_i} + \left(\frac{1}{R_i} \left(\frac{\partial C_i}{\partial SOC} \frac{\partial SOC}{\partial t} + \frac{\partial C_i}{\partial T} \frac{\partial T}{\partial t} \right) \right) \right) U_i + \frac{1}{C_i} I_L, \quad i = 1, \dots, n, \quad (1)$$

$$\frac{dT}{dt} = -\frac{1}{R_T C_T} (T - T_a) + \frac{1}{C_T} Q, \quad (2)$$

$$\frac{dSOC}{dt} = \frac{1}{C_Q} I_L, \quad (3)$$

$$U_L = U_{OC} - I_L R_0 - \sum_{i=1}^n U_i, \quad (4)$$

where R_i , C_i , U_i , I_i , $i = 1, \dots, n$ denote the corresponding resistors, capacitors, voltages and currents of each RC -branch. U_{OC} is the open circuit voltage, I_L the load current and U_L the terminal voltage, R_0 is the ohmic resistance of the battery and the current I_P , the voltage U_P and the resistor R_P describe parasitic effects in the battery. T , T_a are the temperature and the ambient temperature, R_T , C_T the convection resistance and the heat capacity respectively, Q the power dissipated inside the cell and SOC the state of charge and C_Q the total charge of the battery.

This system of equations can be rewritten in more general form using the state space approach:

$$\frac{dx}{dt} = f(x, u, p), \quad x(0) = x_0 \quad (5)$$

$$y = g(x, u, p) \quad (6)$$

with $f: \Omega_x \times \Omega_u \times \Omega_p \mapsto \Omega_f \subset \mathbb{R}^{n+2}$ and $g: \Omega_x \times \Omega_u \times \Omega_p \rightarrow \Omega_g \subset \mathbb{R}$, where the manifolds $\Omega_x \subset \mathbb{R}^{n+2}$, $\Omega_u \subset \mathbb{R}^3$, $\Omega_p \subset \mathbb{R}^{2n+2}$ are suitable chosen. Furthermore $u = (I_L, T_a, Q)^T \in \mathbb{R}^3$, $x = (U_1, \dots, U_n, SOC, T)^T \in \mathbb{R}^{n+2}$, $y = U_L \in \mathbb{R}$ and $p = (R_1, \dots, R_n, C_1, \dots, C_n, U_{OC})^T \in \mathbb{R}^{2n+2}$, $p = p(SOC, T)$ denote the input, state-space, output and parameter vector respectively. In this description the parasitic effects are neglected, all other constants not mentioned in the parameter vector p are assumed to be constant.

This general model for a battery cell is now the starting point for the modeling of the battery packs. Battery packs are included in the description of battery management systems for our development environment. With respect to the high voltage used for electric drives a large amount of cells has to be taken into account for the simulations. This requires accurate but also quick evaluable models. The main disadvantage of the equations given above is the nonlinear character of eq. (1) due to its dependence of the parameters from the SOC and T . Therefore the simplification of the eq. (1) to a linear differential equation is favourable with respect to the parameter identification of the passive electric parts of the network. The major problem of this approach is that the nonlinear functions that describe the dependency of the ECM parameters with respect to cell temperature and SOC are not explicitly known. This problem can be solved by using lookup-tables for the passive elements of the ECM. Lookup-tables (LUTs) are able to approximate these functions via interpolation and extrapolation in the simulation task.

To fill the LUTs several CIT (measurements have been performed in an ARC (ES-ARC, Thermal Hazard Technology) combined with a battery cycler (GSM, BaSyTec GmbH) at different environmental temperatures under isoperibolic conditions. The cell under consideration was a commercial 18650 cell with LiMn_2O_4 cathode, a nominal cell capacity 1600 mAh and a nominal cell voltage of 3.7 V. The calorimeter chamber has several heaters and thermocouples located in the lid, bottom and side walls which adjust the required temperatures depending on the measurement conditions (s. Fig.3). Under isoperibolic conditions, the temperature of the calorimeter chamber is kept constant, whereas the cell temperature can vary.

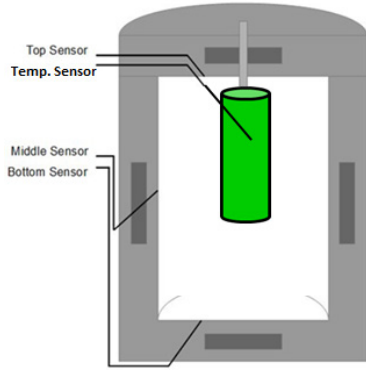


Figure3: Setup of the cell in the ARC.

During the CIT measurement, a cell is first charged to 100% SOC and then discharged by rectangular current pulses of specified amplitude and length (as shown in Fig.4a)), which correspond to a decrease of the SOC of the cell by a well-defined amount. Finally the cell is relaxed over a certain time period. For every single interval of pulse and relaxation the desired parameters of the cell can be identified from the voltage profile (s. Fig.4b)).

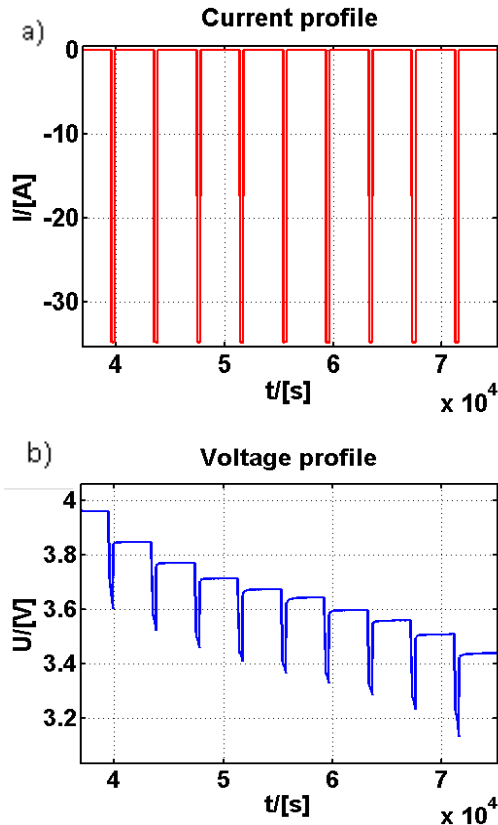


Figure4: CIT measurement: a) applied current profile
b) resulting voltage profile.

From the right end points of every interval one can determine the open-circuit voltage of the cell.

The inner resistance can be extracted from the voltage drop after switching on a discharge pulse. From the relaxation part of the interval one can identify the dominant electro-chemical mechanisms. This parameter identification procedure has been implemented in Matlab[®]. For more details see [13-15].

For the thermal model it was distinguished between the active battery material and the battery can. This thermal model consists of the coupled ordinary differential equations for the temperature of the active material of the cell T_{cell} and the temperature at the cell surface T_{surf} :

$$\frac{dT_{cell}}{dt} = -\frac{1}{R_{T,1}C_{T,1}}(T_{cell} - T_{surf}) + \frac{1}{C_{T,1}}Q, \quad (7)$$

$$\frac{dT_{surf}}{dt} = -\frac{1}{R_{T,2}C_{T,2}}(T_{surf} - T_a) + \frac{1}{R_{T,3}C_{T,3}}(T_{surf} - T_{cell}) \quad (8)$$

The thermal data have been determined in the ARC as described in [15]. The derived LUTs for the electrical and thermal parameters have then been implemented in the next level, which is the virtual design platform.

3 Optimization of the cell model

The parameter values in the look-up tables rely on experimental measurements and are affected with the following errors that have to be taken into account:

- Noise from the measurement,
- Numerical error from the identification procedure,
- Systematic error from measurement strategy and theory behind,
- Errors from uncertainties.

Thus the parameters identified in the last section are in general not the optimal parameters and the look-up tables for the parameters must be optimized. For this purpose an optimization procedure was developed and implemented in Matlab[®]. The general optimization scheme is shown in Fig.5. The purpose of this approach is to eliminate the influence of the different errors to the identified parameters as much as possible. For every temperature T and state of charge SOC the experimental measurement and the simulated model have to be compared. First we describe a single iteration step which is the core of the optimization procedure.

General optimization scheme

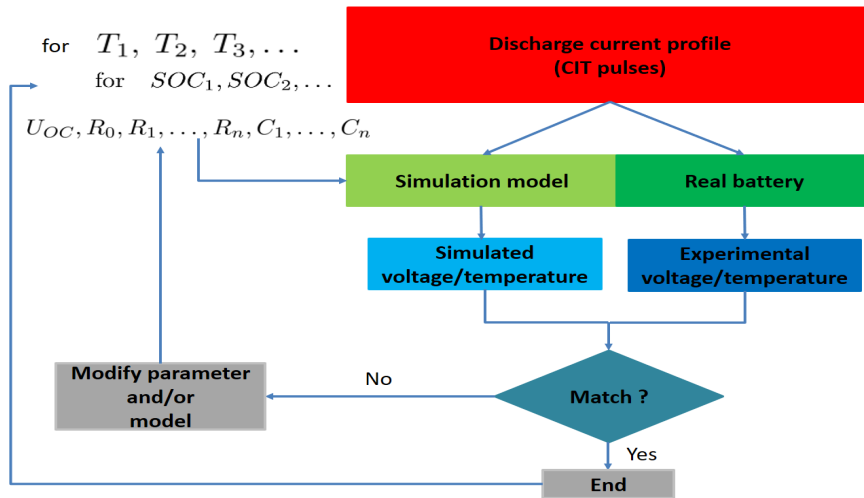


Figure5: General optimization scheme.

4 Simulation

The basis of the virtual design platform is the design environment COSIDE® [16] for the development of electronic systems. COSIDE® is a tool that can be used to model and simulate highly complex electronic and heterogeneous systems and the software running on those systems. To this end, the hardware components are modelled with the hardware description languages SystemC and SystemC AMS. This approach allows for taking the characteristics of the batteries, circuits to balance unequal states of charge (charge balancing), and load profiles into account. In addition, it enables a detailed integration of the components of the battery management integrated circuits (BM ICs) with respect to the characteristics of the A/D converters, delay times etc. and the data interface between BM ICs, and a battery management controller at a high simulation speed.

The ECM of the Li-ion cell has been implemented at the electrical linear network (ELN) level using SystemC AMS as shown in Fig.6a) for the single cell. In addition the model has been extended for a battery pack that consists of six cells in series (6s1p, s. Fig.6b)). In order to ensure a quick model evaluation, a special technique was developed that yields in each time step a linear problem using SOC and temperature from the previous simulation point. The LUTs that have been created from the CIT measurements then approximate the real behaviour by interpolation and extrapolation

during the simulation of the BMS. The model and the parameters are the same for all cells in the pack, they differ only (slightly) in their nominal capacity and in their initial SOC.

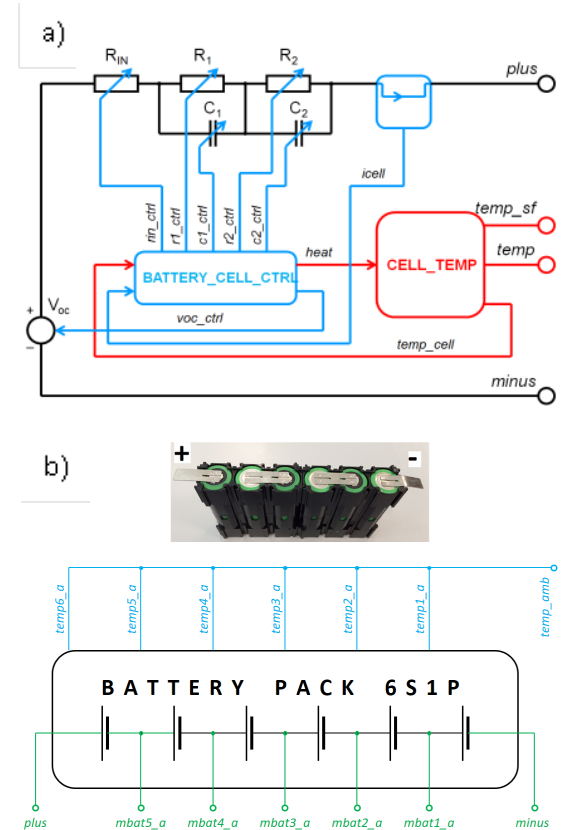


Figure6: Model structure as implemented in SystemC AMS: a) Single cell b) Battery pack 6s1p.

5 Validation of the cell model

For the validation at first the CIT measurements were compared with the simulation results and at second a current profile based on the NEDC (New European Driving Cycle) was applied to make a comparison with the simulation results for a single cell as well as for the battery pack. These measurements were carried out on the cell and the pack in a temperature chamber (Voetsch VT3 4018) with a battery cycler (BaSyTec HPS). The NEDC consists of periods which represent acceleration and periods which represent the slow-down of an electric vehicle. The acceleration and slow-down periods are realized with discharging/charging pulses with different length and amplitude. In total the duration of the cycle is 500s and the decrease in the SOC is $\approx 15\%$. The NEDC was applied to the cell for five different initial state of charge $SOC_{init} = \{20\%, 40\%, 60\%, 80\%, 100\%\}$. Finally in a sixth measurement/simulation a fully charged cell was almost fully discharged by applying the NEDC multiple times. In all three cases a good quantitative agreement has been found as can be seen in Fig.7a) for the single cell at $SOC_{init} = 60\%$ and in Fig.7b) for the battery pack, where an inner and an outer cell have been chosen. When the temperature curves for the single cell are compared, the deviation was only 0.5 K, which is less than the usual 1 K accuracy of the battery temperature measurement on board of an electric vehicle. The heat generations as well as the heat dissipation into the environment were captured

by the model. Thus the presented approach is a good choice for BMS simulations in the design process.

6 Conclusion and outlook

Powerful battery models are central components of simulation environments for the design of battery management systems. Positive experiences were collected with the parametrization of these models on the basis of CIT measurements at different constant temperatures. The parameterization process was carried out in two stages. First, parameters were derived based on terminal currents and voltages of a battery cell with an algorithm for the adaptation of curve trajectories. The values thus determined serve as support points for look-up table models. In a second step, the function values at the grid points were modified in order to achieve a better match between simulation and measurement results for longer measurement sequences. This modification was effected by reducing the deviation as a result of predetermined variations of the functional values at the grid points. The table models thus determined show a good agreement between measurement and simulation results for current profiles, e.g. driving cycles, other than those used in the CIT measurements.

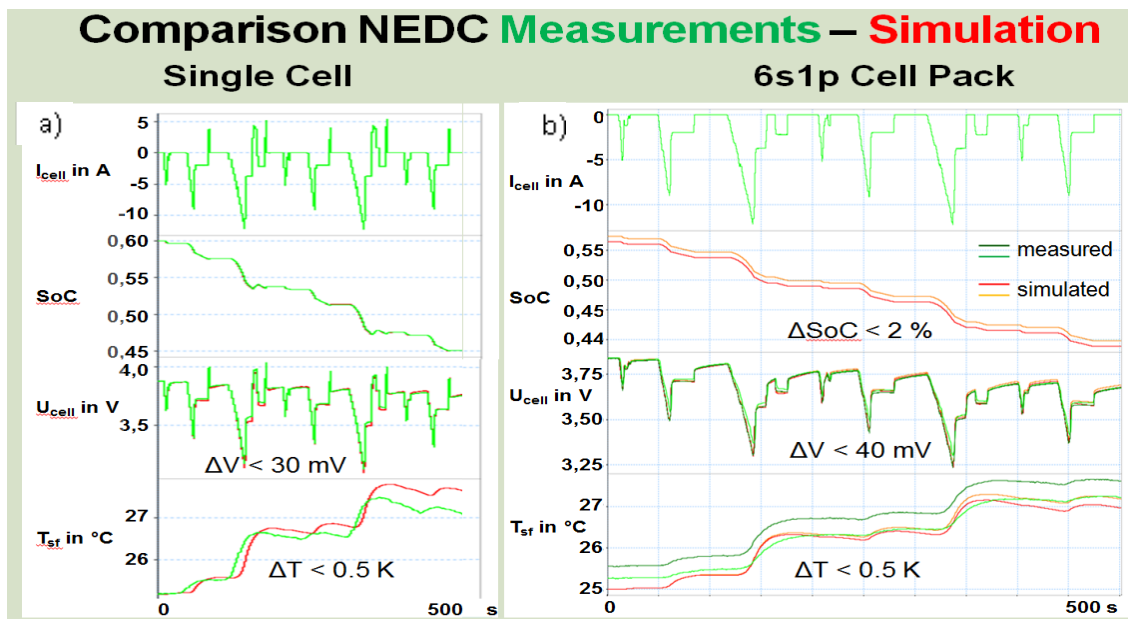


Figure7: Comparison between measurements and simulation for NEDC: a) Single cell

b) Battery pack 6s1p.

Regarding future work, possibly faulty behavior of the components resulting from thermal abuse or other influences in the model will be included. In addition the approach allows for a transition to real-time simulation hardware and thus application in hardware-in-the-loop testing.

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Carlos Ziebert earned his PhD 2002 in Physics from the Saarland University, Germany and joined the KIT. He has more than 15 years of international R&D experience in thin film technology, modelling and thermal characterisation of materials for energy systems and more than 50 peer-reviewed articles. He was leader in 5 projects related to lithium-ion batteries. Currently Dr. Ziebert is head of the KIT Battery Calorimeter Center.



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